



Question Set

Question 1

A two-asset mean-variance investor maximizes

$$\mu^\top w - \lambda w^\top \Sigma w$$

subject to

$$\mathbf{1}^\top w = 1$$

Derive the first-order condition and solve for w in terms of the Lagrange multiplier.

Question 2

For

$$\Sigma = \begin{bmatrix} 0.04 & 0.018 \\ 0.018 & 0.09 \end{bmatrix}, \quad \mu = (0.06, 0.10)^\top, \quad \lambda = 2$$

compute the unconstrained mean-variance stationary portfolio with

$$\mathbf{1}^\top w = 1$$

Question 3

Prove that

$$w^\top \Sigma w$$

is convex when

$$\Sigma$$

is positive semidefinite.

Question 4

Minimize

$$f(x, y) = x^2 + 4xy + 5y^2 - 2x$$

under no constraints. Classify the stationary point using the Hessian.

Question 5

State the KKT conditions for minimizing

$$f(x)$$

subject to

$$g_i(x) \leq 0, \quad h_j(x) = 0$$

Explain each condition in finance language.

Question 6

A trader maximizes log utility

$$\mathbb{E}[\log(1 + wR)]$$

for a small risky allocation w . Use a second-order approximation to derive the approximate optimal w .

Question 7

A nonlinear optimizer returns a solution at a VaR limit boundary. What checks establish whether the boundary solution is economically credible?

Question 8

Explain the difference between local and global minima, and give a calibration example where this matters.

Question 9

For $f(x,y)=(x-1)^4+(y+2)^2$, explain why the Hessian test is inconclusive at part of the optimum but the point is still a minimum.

Question 10

Derive the minimum-variance hedge ratio for minimizing

$$\text{Var}(P - hF)$$

Question 11

What does the Lagrange multiplier on a portfolio budget constraint mean?

Question 12

Why do quant interviewers ask whether the covariance matrix is positive semidefinite before discussing optimization?